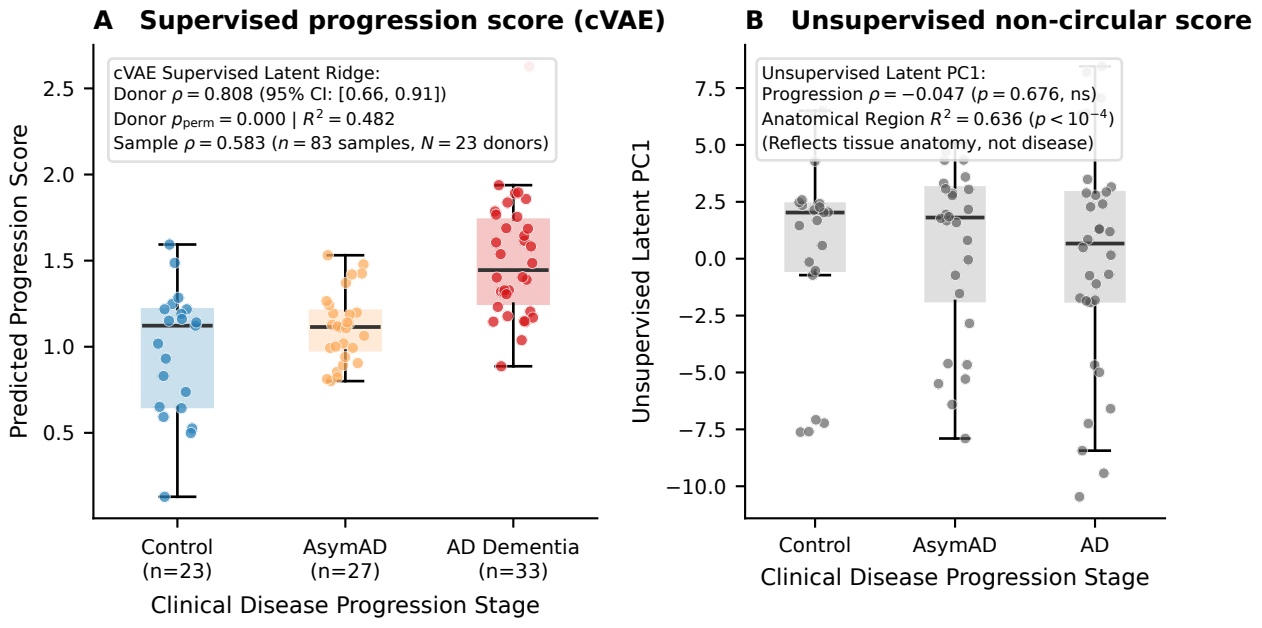
A Supervised progression score (cVAE)
cVAE Supervised Latent Ridge:
Donor $\rho = 0.808$ (95% CI: [0.66, 0.91])
Donor $p_{perm} = 0.000$ | $R^2 = 0.482$
Sample $\rho = 0.583$ ($n = 83$ samples, $N = 23$ donors)
Predicted Progression Score
2.5
2.0
1.5
1.0
0.5
Control (n=23)
AsymAD (n=27)
AD Dementia (n=33)
Clinical Disease Progression Stage
B Unsupervised non-circular score
Unsupervised Latent PC1:
Progression $\rho = -0.047$ ($p = 0.676$, ns)
Anatomical Region $R^2 = 0.636$ ($p < 10^{-4}$)
(Reflects tissue anatomy, not disease)
Unsupervised Latent PC1
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Control
AsymAD
AD
Clinical Disease Progression Stage